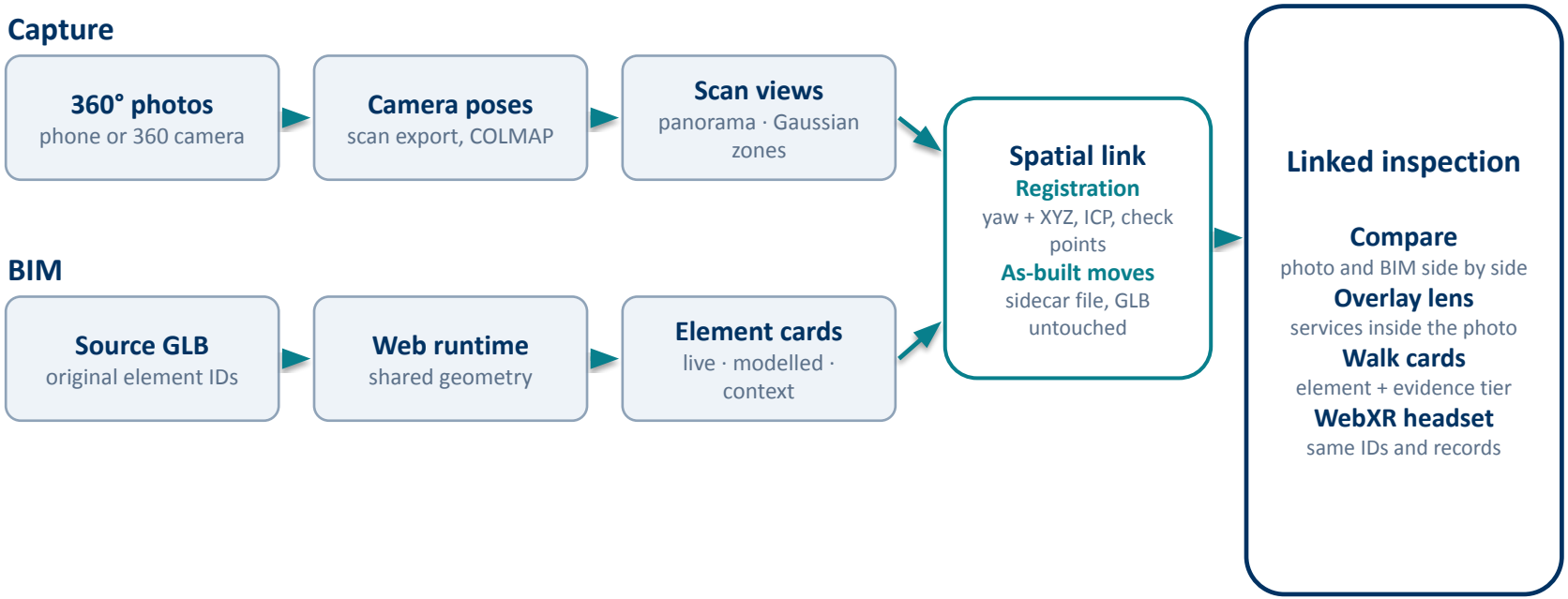
04

Overlay: linking scan and BIM

Services inside the real photo, and how far we can trust it

Implemented research workflow

Capture and BIM meet through a stored, versioned spatial relationship.



Each link keeps its source, version and validation status.

Scan alignment, object identity and sensor mapping keep separate evidence. Gaussian training: Brush; zones let several people train one floor.

Inspect services from the recorded viewpoint

BIM services drawn over the captured 360° photo from the same station.

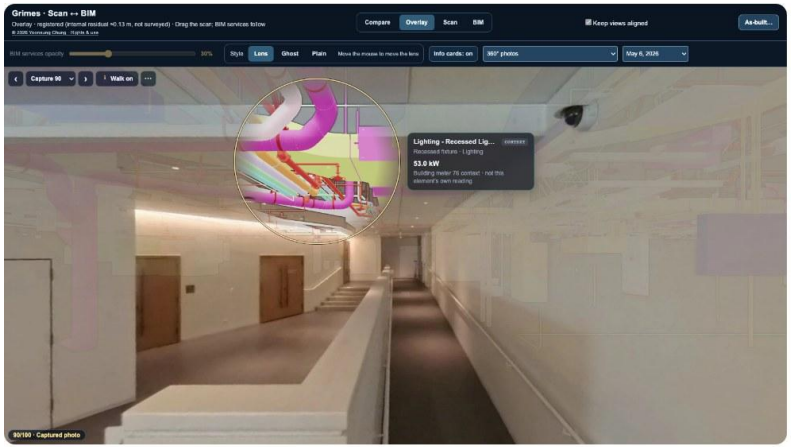


HVAC, lighting and fire protection keep separate source IDs and properties.

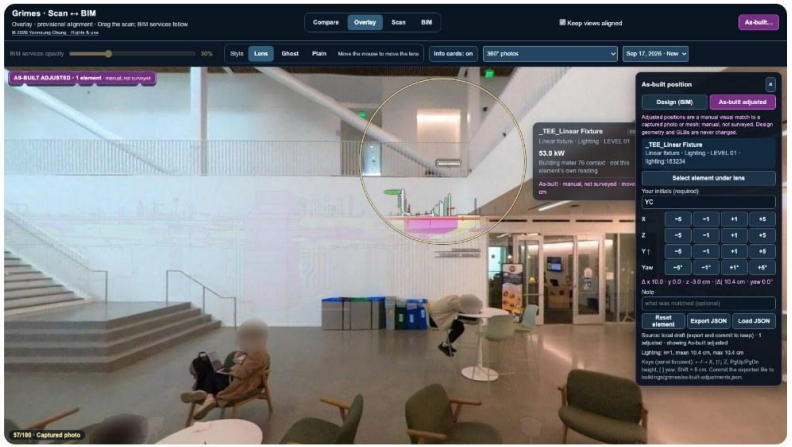
May 6 capture, station 41 · Overlay view (Plain) in scan-compare.html · registration internal residual ≈0.13 m, not surveyed.

X-ray lens with element cards

Services appear only inside the lens; the element under it shows its evidence tier.



Lens + card · May 6, station 90



As-built panel · Sep 17, station 57 (moved 10 cm)

Card: name · system · level · tier (CONTEXT = building meter, not the element's own reading). As-built panel nudges an element to where it was installed.

Cut-away lens (Schall et al. 2009) and focus+context (Kalkofen et al. 2007). As-built moves are manual and not surveyed; Sep 17 uses the provisional alignment.

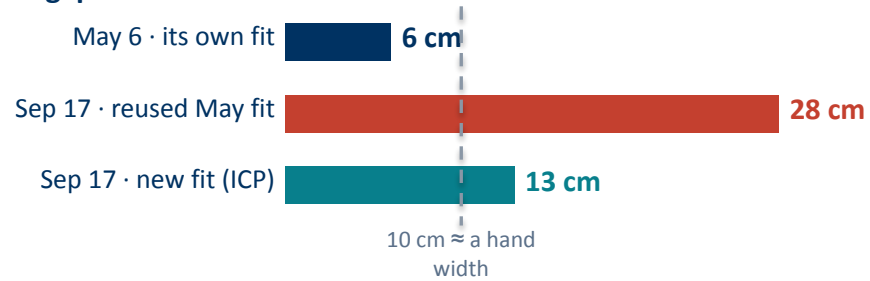
How we correct alignment

Overlaying shows every mismatch, so we measure it, refit, and check it.

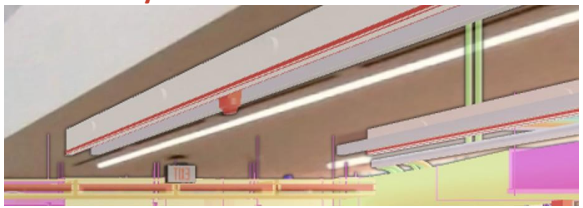
What we measure



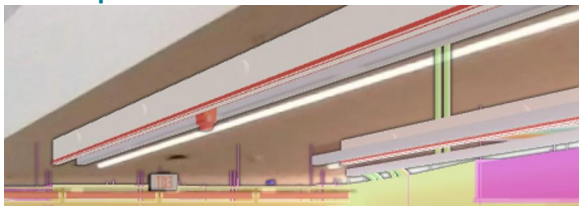
Typical gap at the walls



Reused May fit



New September fit



1 Find the cause
photo direction is fine (≤0.4°), so the fit itself is off

2 Refit
4+ matching points, or the live Align tool

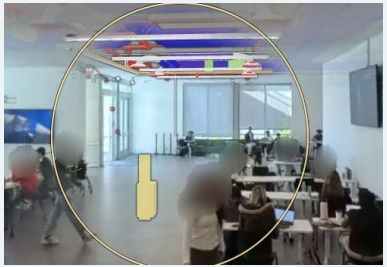
3 Check
points not used in the fit; save as a dated candidate

Refitting September halves the gap (28 → 13 cm); accuracy claims wait for surveyed check points.

Gap = median distance between scan walls and BIM walls along camera rays, per station; internal consistency, not surveyed. Photos: Sep 17, station 56.

What still breaks, and what we are fixing

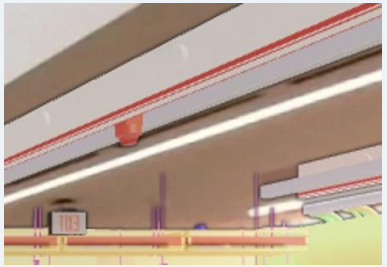
The overlay works today; four gaps stand between it and field AR.



1 No depth in a 360° photo

BIM cannot hide behind real walls or floors, so elements behind a wall or under the slab show through.

Fixing: BIM walls and floor as invisible occluders; ceilings stay see-through.



2 Alignment is not surveyed

Wall offset 0.06 m (May) to 0.13–0.28 m (Sep); the overlay makes small errors visible.

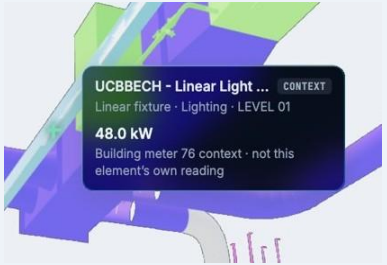
Fixing: Align tool: move, rotate or match elements; then measured check points.



3 BIM is design, not as-built

An element can sit ~10 cm from where it is drawn even when the alignment is right.

Fixing: Per-element as-built moves in a sidecar file; source model untouched.



4 No element sensors yet

Values are building-meter context or modelled shares, labelled as such.

Fixing: Bind panel or BAS points to element IDs so a card can read LIVE.

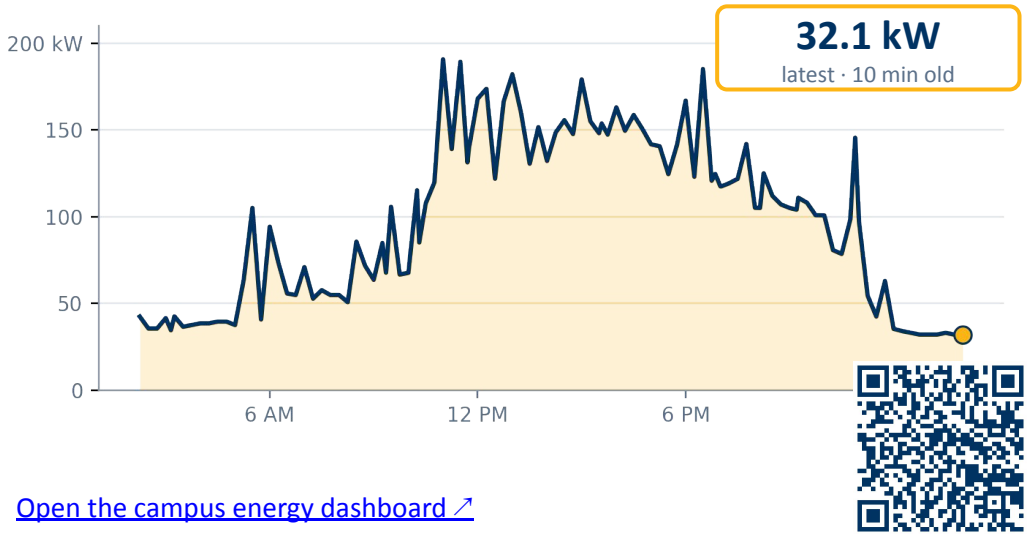
Accuracy stays labelled internal and unsurveyed until independent check points are measured.

3D mesh view retired (reconstruction too distorted); Gaussian zones replace it. Screenshots from the public Scan ↔ BIM page.

Live energy data behind the cards

Grimes meters update every 15 minutes; the same feed gives the cards their context values.

Grimes electric demand, last 24 hours



Open the campus energy dashboard ↗

650 MWh/yr
measured, annualised

0.79×
of the LEED design model (825)

0.43×
of the ASHRAE code baseline

82 / 110
LEED v4 Platinum

How a card labels a number

LIVE	an element sensor (none connected yet)
CONTEXT	the building meter, shown beside an element
MODELLED	a share estimated from design loads

Measured use is 21% below the LEED design model, and cards say when a number is the building meter, not the element.

Source: UC Berkeley BMO meters (device IDs 3, 76, 77), last 24 h to Oct 15 2:00 AM PDT; annual figures 2026-04-11 to 2026-10-14 (187 days) vs LEED v4 EAp2 model.
Dashboards: campus-energy-dashboard.html, grimes-performance.html.